

Errata to “The Universality of Thomas L. Etter’s Three-Place Identity: A Formal Proof and Its Significance”

James Bowery

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The following corrections result from checking the paper’s claims against Etter’s primary sources and the accompanying Lean 4 development. They concern Appendices A and B (the Stereo and RCV constructions) and the Lean modules that formalize them. *The main body of the paper is unaffected.* Definitions D1 and D2, Theorems 4.1–4.3, and their mechanization (the round-trip theorem of §5) are a faithful transcription of Etter’s construction in *Three-place Identity* [E3] and stand unchanged; so does the paper’s use of “universality” in the sense of expressive adequacy / relative interpretability, which is Etter’s own rigorous notion of an open system and is the correct claim.

Erratum 1 — The derived membership of Appendices A and B is not a ZF membership

Location. Definition A.2; the RCV membership of §B.3.

As published. $y \in' x \iff \neg(E_1(y, x) \wedge E_2(y, x))$ (A.2), and $y \in_{\text{RCV}} x \iff \neg(R(y, x) \wedge C(y, x))$ (§B.3), asserted to satisfy / support the axioms of ZF set theory.

Correction. Because E_1, E_2 (resp. R, C) are equivalence relations, hence symmetric, each defined relation is symmetric:

$$y \in' x \iff x \in' y.$$

ZF membership is not symmetric — e.g. $\emptyset \in \{\emptyset\}$ but $\{\emptyset\} \notin \emptyset$ — so neither relation can satisfy Extensionality (or Foundation beyond the trivial $x \notin' x$). Consequently the claims in §A.2–A.3 that this membership yields “full ZF set theory” and in §B.3 that the RCV definition “satisfies the axioms of ZF set theory” are **withdrawn**. Equivalently, non-membership $E_1 \wedge E_2$ is the intersection of two equivalence relations and is therefore itself an equivalence relation — a fact Etter states directly (“only conjunction creates equalities out of equalities”, [E5]).

Separately, the ordered-pair encoding in §A.3 uses a distinct “link” construction that is not derivable from the A.2 membership; the two are independent and should not be presented as a single mechanism.

Erratum 2 — Provenance, and the correct RCV construction

As published. The intersection-style definitions above are presented as Etter’s stereo / RCV membership.

Correction. These formulas are the present author’s reconstructions; they do not occur in Etter’s surviving papers. Etter’s $\text{RCV} \rightarrow \text{ZF}$ membership is the *cell construction* of *The Expressive Power of Equality* [E1]: a cell is an ordered triple $\langle x, y, n \rangle$ with $x \varepsilon y$ and $n \in \{1, 2\}$; its value is $\text{Val}(\langle x, y, n \rangle) = x$ if $n = 1$, else y ; the predicates R, C, V are equalities on cells (same row, same column, same value), with V taken as identity; and membership is recovered from R, C, V alone by

$$\text{First}(c) \iff \forall d \exists e (V(e=d) \wedge C(e=c)),$$

$$c E d \iff \exists e, f (R(e=f) \wedge \text{First}(e) \wedge V(c=e) \wedge \neg \text{First}(f) \wedge V(d=f)).$$

Appendix B should cite [E1] and use this construction. The argument-asymmetry supplied by $\text{First}/\neg\text{First}$ is precisely what the symmetric intersection formula of Erratum 1 lacks.

What stands. D1, D2, Theorems 4.1–4.3, and the *statements* of the stereo and RCV theorems are Etter’s and are transcribed faithfully.

Erratum 3 — Status of the appendix theorems

Correction. The three-equality result — Theorem B.4, that ZF is constructively definable in an RCV language — is not merely asserted. It is proved in full in Etter’s *The Expressive Power of Equality* [E1] by the cell construction above, which also yields the pairing direction (B.5). The only genuinely incomplete item is the two-equality “Stereo Equality Theorem” of Appendix A.

The two-equality theorem is an open conjecture, not merely “stated without proof”. Checking the source closes the gap left in earlier statements of this erratum. Etter announces the theorem in *Three-place Identity* [E3, §3] and defers its proof to Section 3 of *Membership and Identity* [E2]. The surviving M&I draft develops the requisite machinery (intrinsic identity, the notion of a *division*, and the category of *Other*) and then *breaks off before stating or proving the theorem*: its Section 3 ends with a bare, unproved theorem (“the intrinsic identity of a single-place predicate is a division”), Section 4 is an outline, and R. Shoup notes within the draft that the paper was never completed. The only concrete construction Etter gives is the *link sketch* of [E3, fn. 5]: an equality *without transitivity* $[x=y]$ (reflexive and symmetric only), a “meet” operator $[x=:y, z, \dots]$ (“ x equals each of $y, z, \dots$, and anything unequal to y is unequal to x , ditto z ”), and the link membership

$$x \in' y := \exists x', x'', y', y'', q, q'. [q=:x, q'] \ \& \ [q'=y, q, q''] \ \& \ [x'=x, x''] \ \& \ [x''=:x, x'] \ \& \ [y'=y, y''] \ \& \ [y''=:y, y'],$$

with the claim that “there is a set model of $[x=y]$ such that $x \in' y$ is membership in a set model of set theory.” Accordingly the two-equality theorem should be characterised as an *open conjecture* grounded in this sketch, not as an established but unwritten theorem.

Erratum 4 — The Lean formalization of the appendices

Location. The modules `Stereo.lean` and `RCV.lean` accompanying Appendices A and B.

As published. The paper presents a Lean 4 mechanization of the development.

Correction. As published, these two modules do not certify the appendices’ claims. They define `StereoMem` $yx := \neg(E_1 yx \wedge E_2 yx)$ and `Mem_RCV` by projecting R, C onto it (with V unused), and prove only: stereo irreflexivity ($x \notin' x$, trivial), a definitional link lemma, and partial first-component uniqueness. Every such theorem is true, but (i) the membership predicate and the link predicate are never related by any theorem; (ii) no ZF axiom beyond the irreflexivity sliver is established; (iii) the *symmetry* of the defined membership — the property that refutes it as a ZF membership (Erratum 1) — is never stated; and (iv) Etter’s cell apparatus (`cells`, `Val`, `First`, Theorems 1–2 of [E1]) is absent. The modules therefore formalize the incorrect intersection formula and verify only disconnected fragments; they should not be cited as evidence for Appendix B. The main-body module mechanizing Theorem 4.2 (the round-trip) is unaffected and stands.

Corrected development. A faithful mechanization now exists in the accompanying repository [E6]: `Cell.lean` formalizes Etter’s cell construction over Mathlib’s `ZFSet`, discharging Theorems 1 and 2 (`first_iff`, `mem_iff`) with a clean `#print axioms`; `Stereo.Refutation` proves the symmetry that refutes the intersection membership; and the two-equality result is recorded as a precise, falsifiable proposition `etterStereoConjecture`, transcribing the [E3, fn. 5] link construction, rather than as a proved theorem.

Left unchanged (to prevent over-correction)

- “Universality”/“Universality Theorem” and the abstract’s claim of expressive adequacy: Etter’s own rigorous notion of an open/universal system (relative interpretability). Correct; no change.
- Definitions D1, D2 and Theorems 4.1–4.3, with the round-trip Lean proof: faithful and correct.
- The proof of Theorem A.3 (Stereo Irreflexivity): valid *for* Definition A.2. The definition, not the derivation, is at issue in Erratum 1.

Net effect. These corrections *raise* the standing of the underlying program: Etter’s three-equality $\text{RCV} \rightarrow \text{ZF}$ result is complete and correct [E1]. The paper’s appendices substituted a symmetric — hence non-ZF — summary, and a Lean development that certifies only fragments of it, for a proof that already exists and should be cited; and the two-equality claim is more precisely an open conjecture grounded in the [E3, fn. 5] link sketch.

References

- [E1] Etter T. *The Expressive Power of Equality*. Boundary Institute; June 20, 2001. (Contains the cell construction and the proof of the expressiveness/RCV theorem.)
- [E2] Etter T. *Membership and Identity*. Boundary Institute; 2006. (Incomplete draft.)
- [E3] Etter T. *Three-place Identity (Set Theory and the Identity of Mathematical Objects)*. Boundary Institute; 2006.
- [E4] Etter T. *Notes on Identity and Pairing*. Boundary Institute; 2001.

- [E5] Etter T. *Equalities and Quine Identities*, §§1–6. Boundary Institute; 2001.
- [E6] Bowery J. *RelativeIdentity* (Lean 4 sources; `Cell.lean`, `RCV.lean`, `Stereo.lean`). <https://github.com/jabowery/RelativeIdentity>